

Prime and Non-Prime Attributes in DBMS

Last Updated: August 11, 2025

Definition Overview

Prime Attribute: An attribute that is part of at least one *candidate key*.

Non-Prime Attribute: An attribute that is not part of any candidate key.

Understanding the Concepts

Candidate Key Recap

A **candidate key** is a minimal set of attributes that can uniquely identify each tuple in a relation.

Candidate Key $\implies$ Uniqueness + Minimality

If an attribute appears in *any* candidate key, it is **prime**; otherwise, it is **non-prime**.

Visual Example

Consider the relation:

$R(A, B, C, D)$

with functional dependencies:

$A \rightarrow B, \quad C \rightarrow D, \quad AC \rightarrow B$

and suppose:

Candidate Keys: $\{A, C\}, \{B, C\}$

- **Prime Attributes:** A, B, C (appear in at least one candidate key)
- **Non-Prime Attributes:** D (not in any candidate key)

How to Determine Prime and Non-Prime Attributes

Step-by-Step Method

1. List all Functional Dependencies (FDs).
2. Find all Candidate Keys:

- Compute attribute closures.
- Identify minimal attribute sets with full closure = all attributes.

3. Mark Prime Attributes:

- If an attribute is in *any* candidate key, label it as **prime**.

4. Remaining attributes are Non-Prime.

Example Walkthrough

Given:

$R(P, Q, R, S)$, FDs: $P \rightarrow Q$, $Q \rightarrow R$, $PR \rightarrow S$

1. Find Candidate Keys:

- $P^+ = \{P, Q, R\}$ (not all attributes)
- $(P, R)^+ = \{P, Q, R, S\} \rightarrow$ Candidate Key

2. Prime Attributes: P, R

3. Non-Prime Attributes: Q, S

Importance in Normalization

- Helps identify **partial** and **transitive** dependencies.
- Decides violation of **2NF** and **3NF**:
 - **2NF**: No partial dependency of non-prime attribute on candidate key.
 - **3NF**: Every non-prime attribute must be fully dependent on candidate key and only on candidate key.

Summary Table

Attribute Type	Definition
Prime	Appears in at least one candidate key
Non-Prime	Appears in no candidate key